When sitting at rest in the laboratory, muons disintegrate by a process closely akin to radioactive decay, in an average of about two millionths of a second. This disintegration is an experimental fact supported by an enormous amount of evidence. It's as if a muon lives its life with a gun to its head; when it reaches two millionths of a second in age, it pulls the trigger and explodes apart into electrons and neutrinos. But if these muons are not sitting at rest in the laboratory and instead are traveling through a piece of equipment known as a particle accelerator that boosts them to just shy of light-speed, their average life expectancy as measured by scientists in the laboratory increases dramatically. This really happens. At 667 million miles per hour (about 99.5 percent of light speed), the muon lifetime is seen to increase by a factor of about ten. The explanation, according to special relativity, is that "wristwatches" worn by the muons tick much more slowly than the clocks in the laboratory, so long after the laboratory clocks say that the muons should have pulled their triggers and exploded, the watches on the fast-moving muons have yet to reach doom time. This is a very direct and dramatic demonstration of the effect of motion on the passage of time. If people were to zip around as quickly as these muons, their life expectancy would also increase by the same factor. Rather than living seventy years, people would live 700 years.3

Now for the catch. Although laboratory observers see fast-moving muons living far longer than their stationary brethren, this is due to time elapsing more slowly for the muons in motion. This slowing of time applies not just to the watches worn by the muons but also to all activities they might undertake. For instance, if a stationary muon can read 100 books in its short lifetime, its fast-moving cousin will also be able to read the same 100 books, because although it appears to live longer than the stationary muon, its rate of reading—as well as everything else in its life—has slowed down as well. From the laboratory perspective, it's as if the moving muon is living its life in slow motion; from this viewpoint the moving muon will live longer than a stationary one, but the "amount of life" the muon will experience is precisely the same. The same conclusion, of course, holds true for the fast-moving people with a life expectancy of centuries. From their perspective, it's life as usual. From our